

Mask-Path InfoNCE without EMA in a Global Image-Level JEPA: A CIFAR Study

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Abstract

Joint-Embedding Predictive Architectures (JEPAs) commonly stabilize latent prediction with an exponential-moving-average (EMA) target encoder. We test whether applying InfoNCE directly to the masked predictive pair can instead train a single ResNet-18 encoder with stop-gradient targets and no EMA. Under a matched 100-epoch CIFAR-10 protocol across three seeds, mask-path InfoNCE reaches $42.6 \pm 3.6\%$ k -nearest-neighbor accuracy and $43.0 \pm 3.4\%$ linear-probe accuracy, compared with $41.6 \pm 2.9\%$ and $41.3 \pm 2.7\%$ for cosine JEPA with an EMA target. The no-EMA method has slightly higher means, although the observed variability does not establish a decisive advantage. These results identify mask-path contrast as a viable replacement for EMA stabilization in this setting.

1 Introduction

Self-supervised learning (SSL) for visual representations spans two influential families: *contrastive* instance discrimination [4, 8] and *predictive* joint embedding (JEPA) [1, 10]. I-JEPA predicts target-block embeddings from a context encoder and uses regression in latent space with an EMA target network. C-JEPA [11] combines I-JEPA with variance-invariance-covariance regularization (VICReg) [3] to stabilize training and improve convergence on ImageNet. These methods suggest several plausible substitutes for an EMA target: batch negatives, variance-covariance constraints, or distribution matching. However, a JEPA pipeline with both masked and augmented views introduces a placement choice: the anti-collapse signal can act on masked prediction or on a separate augmentation path.

We therefore ask: *for a single-encoder JEPA without EMA, should contrast act directly on masked*

prediction, or should masked prediction remain non-contrastive while a separate augmentation path prevents collapse? We compare cosine mask→clean prediction plus augmentation→clean InfoNCE with mask-path InfoNCE and non-contrastive augmentation regularizers.

Contributions. (i) A controlled formulation that separates the masked predictive path from the augmentation anti-collapse path; (ii) a matched CIFAR-10 comparison of contrastive, variance-covariance, and Gaussian regularization with a single ResNet-18 encoder; (iii) evidence that mask-path InfoNCE is the strongest tested no-EMA objective, while augmentation-side regularization does not improve on that baseline.

2 Related work

Predictive JEPA. I-JEPA [1] predicts latent targets under masking with smooth-L1 / L2 and an EMA teacher, following the JEPA agenda of LeCun [10].

C-JEPA. C-JEPA [11] argues that EMA alone is insufficient to prevent collapse and that mean-patch prediction is underconstrained; it augments I-JEPA with VICReg [3] variance / covariance / invariance terms and reports faster ImageNet convergence. Our note shares C-JEPA’s goal of combining predictive and contrastive-style signals, but studies a much smaller CIFAR recipe: one encoder, stop-grad (no EMA), cosine JEPA on the mask path, and either augmentation InfoNCE or a lightweight regularizer (mean squared error with variance and covariance penalties, VICReg, or Sketched Isotropic Gaussian Regularization (SIGReg) [2]).

Contrastive and non-contrastive SSL. SimCLR [4] and MoCo [8] use InfoNCE on two augmented views. BYOL [6] and SimSiam [5] avoid ex-

explicit negatives via predictors and stop-grad / EMA. Barlow Twins [12] and VICReg [3] prevent collapse with redundancy-reduction / variance-covariance penalties rather than negatives.

3 Methods and Objectives

Views and notation. For each image x , we form a lightly augmented clean view x_c , a block-masked view x_m , and a strongly augmented view x_a . A single encoder f_θ maps each view to a 512-dimensional representation, and a predictor g_ϕ is used only on the masked path:

$$z_c = f_\theta(x_c), \quad \hat{z}_m = g_\phi(f_\theta(x_m)), \quad z_a = f_\theta(x_a). \quad (1)$$

The predictor is discarded at evaluation. Except for the EMA reference, all objectives use the same online encoder. The notation $\text{sg}(\cdot)$ denotes stop-gradient. It is applied to the clean target in the cosine and InfoNCE predictive pairs, but not between the two branches of the variance-covariance or SIGReg augmentation regularizers.

Cosine JEPA. The non-contrastive masked-prediction objective is

$$L_{\cos} = 1 - \frac{\hat{z}_m^\top \text{sg}(z_c)}{\|\hat{z}_m\|_2 \|\text{sg}(z_c)\|_2}. \quad (2)$$

Averaging over the batch is implicit. This objective enforces invariance between masked and clean views but contains no explicit term requiring features from different images to remain distinct. Consequently, a single-encoder version can admit collapsed or weak representations. The EMA baseline uses the same cosine loss but computes the clean target with a momentum encoder f_ξ , updated as $\xi \leftarrow m\xi + (1-m)\theta$ with $m = 0.996$.

InfoNCE. For query-key pairs q_i, k_i in a batch of size B , we use

$$L_{\text{NCE}}(q, k) = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp(\bar{q}_i^\top \bar{k}_i / \tau)}{\sum_{j=1}^B \exp(\bar{q}_i^\top \bar{k}_j / \tau)}, \quad (3)$$

where bars denote ℓ_2 normalization and $\tau = 0.1$. The matching view of the same image is the positive; the other $B - 1$ clean embeddings are negatives. Stop-gradient is applied to every key. We compare two placements:

$$L_{\text{mask-NCE}} = L_{\text{NCE}}(\hat{z}_m, \text{sg}(z_c)), \quad (4)$$

$$L_{\cos+\text{aug-NCE}} = L_{\cos} + L_{\text{NCE}}(z_a, \text{sg}(z_c)). \quad (5)$$

Thus, mask-only InfoNCE makes the predictive task itself contrastive, whereas the second objective preserves non-contrastive masked prediction and places instance discrimination only on the augmentation path. Both active terms have unit weight.

MSE, variance, and covariance. To test negative-free anti-collapse regularization, we replace augmentation InfoNCE with the three VICReg building blocks. For $z \in \mathbb{R}^{B \times D}$,

$$L_{\text{inv}} = \frac{1}{BD} \|z_a - z_c\|_F^2, \quad (6)$$

$$v(z) = \frac{1}{D} \sum_{d=1}^D \max\left(0, 1 - \sqrt{\text{Var}(z_{:,d}) + \epsilon}\right), \quad (7)$$

$$c(z) = \frac{1}{D} \sum_{r \neq s} \text{Cov}(z)_{rs}^2, \quad (8)$$

with $\epsilon = 10^{-4}$. The invariance term aligns the two views, the variance hinge prevents individual dimensions from becoming constant, and the covariance penalty discourages redundant dimensions. The complete objective is

$$L_{\text{MVC}} = L_{\cos} + \lambda_i L_{\text{inv}} + \lambda_v [v(z_a) + v(z_c)] + \lambda_c [c(z_a) + c(z_c)]. \quad (9)$$

We evaluate unit weights $(\lambda_i, \lambda_v, \lambda_c) = (1, 1, 1)$ and the VICReg-scale setting (25, 25, 25) reported as separate rows. Gradients flow through both z_a and z_c in these three terms.

SIGReg. Sketched Isotropic Gaussian Regularization (SIGReg) replaces the explicit variance and covariance penalties with a distribution-matching term. We concatenate both augmentation branches, $Z = [z_a; z_c]$, project Z onto 256 deterministically seeded random unit directions, and compare each projected empirical characteristic function with that of $\mathcal{N}(0, 1)$ using the Epps-Pulley statistic [2]. In compact form,

$$L_{\text{SIG}} = L_{\cos} + L_{\text{inv}} + \frac{1}{R} \sum_{r=1}^R \mathcal{E}(Zu_r, \mathcal{N}(0, 1)), \quad R = 256, \quad (10)$$

where u_r is a random unit direction and $\mathcal{E}$ is the weighted characteristic-function discrepancy. All three terms use unit weight. SIGReg encourages a non-degenerate, approximately isotropic latent distribution while L_{inv} preserves augmentation invariance.

Table 1: CIFAR-10, ResNet-18, evaluated at the fixed 100-epoch endpoint. Learned-model results are mean $\pm$ sample standard deviation over seeds 42, 7, and 123. Random chance is included as a reference baseline. Cosine JEPA + augmentation InfoNCE places batch negatives only on the augmentation path. Hybrids = cosine JEPA (mask) + listed augmentation/clean regularizer (no EMA). All reported runs are complete.

Method	k -NN	Linear	Seeds
<i>Reference baselines</i>			
Random classifier (chance)	0.100	0.100	–
Supervised CE	0.949 ± 0.001	0.948 ± 0.001	3
<i>Self-supervised learning</i>			
EMA JEPA cosine	0.416 ± 0.029	0.413 ± 0.027	3
JEPA InfoNCE (mask only; no EMA)	0.426 ± 0.036	0.430 ± 0.034	3
Cosine JEPA + augmentation InfoNCE (no EMA)	0.402 ± 0.010	0.399 ± 0.019	3
JEPA + MSE/variance/covariance (no EMA)	0.397 ± 0.002	0.346 ± 0.063	3
JEPA + VICReg (no EMA; 25/25/25)	0.374 ± 0.004	0.393 ± 0.007	3
JEPA + SIGReg (no EMA)	0.383 ± 0.008	0.407 ± 0.004	3

Table 2: CIFAR-100, ResNet-18, seed 42, evaluated at the fixed 100-epoch endpoint. These exploratory single-seed results are reported separately from the three-seed CIFAR-10 comparison and do not include uncertainty estimates.

Method	k -NN	Linear	Epoch
Random classifier (chance)	0.010	0.010	–
EMA JEPA cosine	0.125	0.129	100
JEPA InfoNCE (mask only; no EMA)	0.155	0.150	100
Cosine JEPA + augmentation InfoNCE (no EMA)	0.118	0.106	100

Controlled comparison. Across the no-EMA rows, the encoder, predictor, views, optimizer, and training budget are fixed; only the objective and its stated weights change. This separation lets us test whether anti-collapse behavior depends not only on the regularizer family but also on the path to which it is applied.

4 Experimental protocol

Data and architecture. CIFAR-10 [9]; ResNet-18 [7] with embedding dimension 512; a predictor multilayer perceptron (MLP) on the mask path. Block masking with mask ratio 0.6. Strong augs: random resized crop, horizontal flip, color jitter ($B=C=S=0.4$, hue 0.1).

Optimization. Adam, learning rate 3×10^{-4} , batch size 256, 100 epochs, InfoNCE temperature $\tau=0.1$, one encoder (no EMA).

Evaluation. Frozen encoder; k -NN ($k=20$) and linear probe (standard CIFAR split). Evaluate every 10 epochs and select the checkpoint with best k -NN.

Hardware and runtime. Experiments were run on NVIDIA GeForce RTX 5060 Ti GPUs (16 GB), using one GPU per run. Including periodic frozen-feature evaluation, the 100-epoch mask-only InfoNCE run took 11,721 s (3.26 h), while cosine JEPA plus augmentation InfoNCE took 10,546 s (2.93 h). The two runs can be executed concurrently on two GPUs, for approximately 3.26 h wall-clock time.

Baselines (same stack).

- Supervised cross-entropy (CE; upper bound);
- EMA JEPA cosine (teacher EMA);
- JEPA InfoNCE on the mask path only (no augmentation term);
- Cosine JEPA + augmentation InfoNCE;
- Hybrids: cosine JEPA (mask→clean, no EMA) + MSE/variance/covariance, SIGReg [2], or VICReg [3] on augmentation/clean;

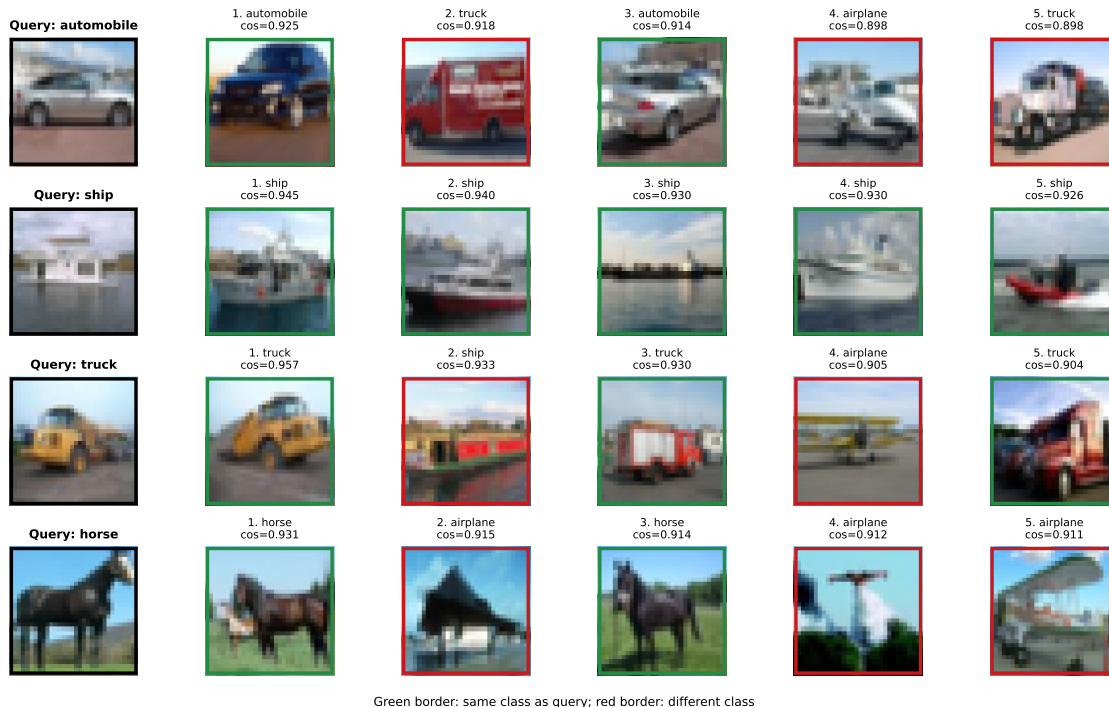


Figure 1: Five nearest training images for four test queries under cosine similarity in the frozen mask-only InfoNCE representation. The encoder is the best checkpoint at epoch 80. The four representative queries span automobile, ship, truck, and horse; their fixed test indices are provided with the figure metadata. Retrieval uses all 50,000 training images with deterministic evaluation preprocessing. Green borders indicate the query class and red borders a different class. The examples expose both coherent neighborhoods and cross-class confusion, consistent with the moderate aggregate k -NN accuracy.

5 Results

All numbers below are CIFAR-10, ResNet-18 ($D=512$), batch 256, 100 epochs, frozen k -NN ($k=20$) and linear probe. We report the fixed epoch-100 endpoint as mean $\pm$ sample standard deviation over three seeds (42, 7, and 123) for both supervised and self-supervised models.

Qualitative neighborhoods. Figure 1 shows the five most similar training images for fixed test queries spanning four classes. The class-consistent ship neighborhood contrasts with mixed automobile, truck, and horse neighborhoods. This qualitative variation helps explain why local similarity is useful but far from class-perfect.

Mask-path contrast is strongest. Mask-only InfoNCE is the best tested SSL method by both k -NN (0.426 ± 0.036) and linear probe (0.430 ± 0.034). It exceeds the EMA cosine-JEPA reference by about

one percentage point on each mean while using no teacher encoder. This suggests that batch negatives on the predictive pair can supply an effective anti-collapse signal in the present regime, although the overlapping uncertainty cautions against claiming a decisive advantage over EMA.

Augmentation-side contrast is not an equivalent substitute. Cosine JEPA plus augmentation InfoNCE reaches 0.402 ± 0.010 k -NN and 0.399 ± 0.019 linear accuracy, trailing mask-only InfoNCE on both means. Its smaller variance does not recover the average representation quality, supporting the view that contrast placement matters.

Distributional regularizers trail mask-path contrast. The cosine-JEPA regularizers do not recover the mask-only InfoNCE result. MSE/variance/covariance has stable k -NN (0.397 ± 0.002) but a variable linear probe (0.346 ± 0.063). VICReg reaches

$0.374 \pm 0.004 / 0.393 \pm 0.007$, while SIGReg reaches $0.383 \pm 0.008 / 0.407 \pm 0.004$ (k -NN/linear). The mismatch between neighborhood and linear metrics cautions against ranking representations with only one evaluator.

6 Discussion and limitations

Relation to C-JEPA. C-JEPA [11] combines I-JEPA with VICReg on large-scale ImageNet pre-training and retains an EMA teacher. We target a minimal CIFAR recipe without EMA: cosine JEPA on the mask path for “ours” and for the regularizer hybrids (MSE/variance/covariance, SIGReg [2], VICReg [3]).

Limitations. CIFAR-10, ResNet-18, and three seeds are insufficient for broad claims about JEPA training, and the objectives were not tuned with equal hyperparameter budgets. We lack a matched augmentation-InfoNCE-only control and ImageNet-scale validation. Although fixed-epoch reporting avoids selecting the endpoint by test k -NN, hyperparameter development still used the same benchmark; future experiments should use a held-out validation split. Finally, our global image-level target simplifies I-JEPA’s block-level prediction task.

Takeaway. A single encoder can learn non-collapsed masked representations without EMA, but the successful ingredient is not merely the presence of an anti-collapse loss. At epoch 100, mask-path InfoNCE is the strongest tested choice on average (0.426 ± 0.036 k -NN, 0.430 ± 0.034 linear); cosine prediction paired with augmentation-side InfoNCE or distributional regularization is weaker under the evaluated settings. The practical lesson is to treat loss placement as a first-class design choice in predictive-contrastive SSL.

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